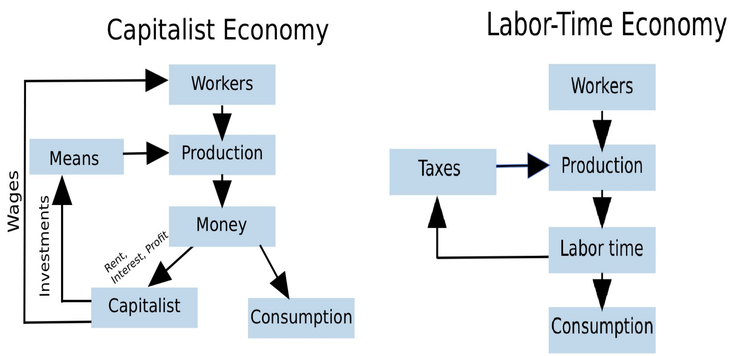
Sharing Productivity Benefits in a Labor-Time Economy

Salamun Nuhin*. Advisors: Alexander Creiner*, Ivan Williams**, Amittai Aviram* (*Boston College, **UMass)

Background

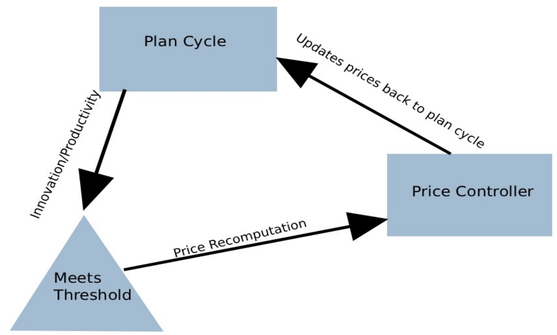


The capitalist productivity paradigm creates a fundamental contradiction: worker innovations primarily benefit owners, while innovative workers face job insecurity through layoffs and work intensification. This creates perverse incentives, where workers rationally resist productivity improvements to preserve employment, asking, "Why should I innovate myself out of a job while enriching others?"

Current economic structures misalign individual worker incentives with collective productivity goals, resulting in suboptimal innovation rates and economic inefficiency.

Through agent-based simulation modeling, this research examines a labor time economy, where workers are paid in labor-time credits, which they turn in for consumption, with no credits diverted to profits. Productivity gains translate directly to workers through decreased labor requirements. while maintaining full compensation, creating positive incentives for continuous improvement and shared prosperity.

Methods

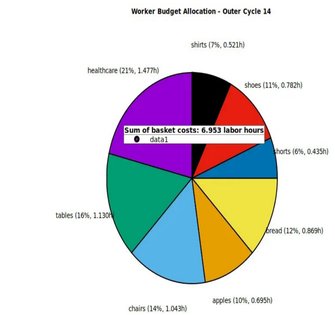


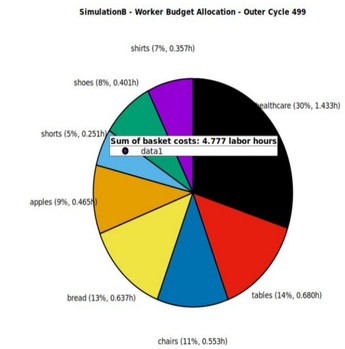
The central challenge is designing economic conditions that encourage productivity improvements without penalizing workers in more efficient sectors. While Toyota's production system shows that workers naturally contribute innovations without direct financial incentives, in a labor-time economy, productivity gains could cause workers to earn fewer credits and lose purchasing power, effectively punishing them.

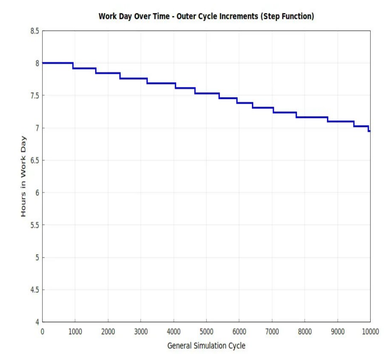
We investigated two solutions: (a) Whenever an innovation reduces required labor time for a product, its price *and* those workers' labor credits are immediately reduced; (b) Workers in a sector with productivity improvements continue to receive credits according to the previously necessary labor time, and prices and pay are readjusted only after enough different sectors have similar productivity gains.

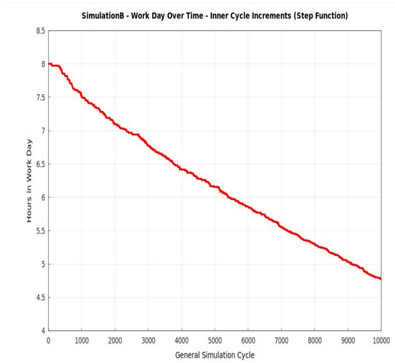
We simulated 8 sectors over 10,000 cycles (27 years), using Toyota production data. Innovation follows Poisson distribution (15% base probability, 1% variance), generating 1-20% productivity improvements, with price recomputation triggered when ≥4 firms achieve ≥4% efficiency gains. We sought to determine which approach maintains both innovation incentives and economic stability while preventing purchasing power disparities between sectors.

Results









The simulation reveals critical differences between controlled and uncontrolled productivity benefit distribution. With immediate price and wage changes (solution (a)), significant disparities emerge. Workers in high-productivity sectors see hours drop from 8.0 to under 5.0, but their budget allocation becomes skewed. Healthcare becomes disproportionately expensive relative to reduced labor time, creating affordability crises for workers in other sectors who have less purchasing power due to fewer hours. Costs for a basket of essentials drop to 4.777 hours, but create a two-tier system where more productive sector workers cannot afford the same living standards, especially for healthcare.

Under delayed price and wage recomputation (solution (b)), workers maintain stable purchasing power as hours decline from 8.0 to 7.0 over 10,000 cycles while budget allocation stays balanced across sectors. Costs for a basket of essentials remain manageable (6.953 hours in cycle 14 versus 8.000 initially), maintaining proportional access to goods.

This validates proposed solution (b). Occasional and coordinated price recomputation: prevents purchasing power divergence, which would undermine a labor-time economy's promise of equality. The controlled mechanism successfully distributes productivity benefits across the workforce rather than ironically concentrating advantages in the least productive sectors.

References

IdeaWake. "The Success of Toyota's Employee Suggestion Program." IdeaWake, 2023, ideawake.com/the-success-of-toyotas-employee-suggestion-program/.

Kesting, Peter, and Jens Kappe. "Getting the Measure of Employee-Driven Innovation and Its Workplace Correlates." ResearchGate, 2020, www.researchgate.net/publication/339584385_Getting_the_Measure_of_Employee-Driven_Innovation_and_Its_Workplace_Correlates_Employee-Driven_Innovation
.

Martinsson, Gustav. "Innovation, Productivity and Patent Quality: Micro Evidence from Swedish Manufacturing." arXiv, 22 Oct. 2022, arxiv.org/abs/2210.04102.

Toivanen, Otto, and Lotta Väänänen. "Innovation and Labor Markets." Science Direct, vol. 27, 2023, www.sciencedirect.com/science/article/abs/pii/S0927539823001020.

Group of International Communists. Fundamental Principles of Communist Production and Distribution. Edited and translated by Hermann Lueer, Independently Published, 2020.

Law, Averill M. Simulation Modeling and Analysis. 5th ed., McGraw-Hill, 2014.

Beer, Stafford. Brain of the Firm. 2nd ed., Wiley, 1995.